**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Bar Vee Dairy, Inc.

Physical address of dairy:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 3031 N Washington RD | Turlock | Stanislaus | 95380 |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1973

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0023-0056-0005-0000 0023-0056-0006-0000

**B. OPERATORS**

Bar Vee, Dairy

Operator name: Bar Vee, Dairy

Telephone no.: (209) 632-4308 (Landline) (Cellular)

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 431 N Washington RD | Turlock | CA | 95380 |

Crowell, Adam

Operator name: Crowell, Adam

Telephone no.: (209) 632-4308 (Landline) (Cellular)

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 431 N Washington RD | Turlock | CA | 95380 |

**This operator is responsible for paying permit fees.**

**C. OWNERS**

Crowell, Adam

Legal owner name: Crowell, Adam

Telephone no.: (209) 632-4308

| Landline | Cellular |
|---|---|
| (209) 632-4308 | |

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 431 N Washington RD | Turlock | CA | 95380 |

**This owner is responsible for paying permit fees.**

AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 0 | 0 | 88 | 80 | 90 | 50 |
| Number under roof | 305 | 43 | 0 | 0 | 0 | 0 |
| Maximum number | 305 | 43 | 88 | 80 | 90 | 50 |
| Average number | 305 | 43 | 88 | 80 | 90 | 50 |
| Avg live weight (lbs) | 1,300 | 1,400 | 900 | 700 | | |

Predominant milk cow breed: Other

Average milk production: 80 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 10,937.36 *tons per reporting period*

Total nitrogen from manure: 137,915.73 *lbs per reporting period*

After ammonia losses (30% loss applied): 96,541.01 *lbs per reporting period*

Total phosphorus from manure: 22,417.62 *lbs per reporting period*

Total potassium from manure: 60,414.16 *lbs per reporting period*

Total salt from manure: 153,497.10 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 2,999,996 *gallons*

Total nitrogen generated: 2,032.84 *lbs*

Total phosphorus generated: 615.86 *lbs*

Total potassium generated: 11,415.94 *lbs*

Total salt generated: 53,574.83 *lbs*

| | | |
|---|---|---|
| | 2,999,996 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 0 | *gallons imported* |
| = | 2,999,996 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| TID canal | Surface water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

*No process wastewater nutrient imports entered.*

| Date | Material type / Description | Quantity | Reporting basis | Moisture (%) | N (%) | P (%) | K (%) | Salt (%) |
|---|---|---|---|---|---|---|---|---|
| 06/07/2023 | Liquid commercial fertilizer<br>*NH3* | 4,609.00 *gal* | | | 82.000000 | 0.000000 | 0.000000 | 0.000000 |
| 08/16/2023 | Liquid commercial fertilizer<br>*UAN 32* | 1,000.00 *gal* | | | 32.000000 | 0.000000 | 0.000000 | 0.000000 |

| Material type | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Commercial fertilizer / Other | 34,209.33 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Total imports for all materials | 34,209.33 | 0.00 | 0.00 | 0.00 |

## G. NUTRIENT EXPORTS

*No solid nutrient exports entered.*

*No liquid nutrient exports entered.*

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| 1 | 30 | 30 | 1 | both | 023X-056X-004X-XXXX<br>023X-056X-005X-XXXX |
| 2 | 80 | 80 | 1 | both | 023X-056X-004X-XXXX<br>023X-056X-005X-XXXX |
| 3 | 80 | 80 | 1 | both | 023X-056X-004X-XXXX<br>023X-056X-005X-XXXX |
| 4 | 20 | 20 | 1 | manure | 087X-003X-018X-XXXX |
| Pasture | 8 | 8 | 0 | none | 023X-056X-005X-XXXX |
| Totals for areas that were used for application | 210 | 210 | 4 | | |
| Totals for areas that were not used for application | 8 | 8 | 0 | | |
| Land application area totals | 218 | 218 | 4 | | |

## B. CROPS AND HARVESTS

1

Field name: 1

04/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 04/06/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 11/30/2023 | 1,200.00 *ton* | Dry-weight | | 72.9 | 12,100.00 | 2,605.00 | 15,700.00 | | 5.43 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 40.00 | 320.00 | 60.00 | 264.00 | 0.00 |
| Total actual harvest content | 40.00 | 262.33 | 56.48 | 340.38 | 1,177.22 |

2

Field name: 2

2

04/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 04/08/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 11/29/2023 | 3,200.00 *ton* | Dry-weight | | 71.0 | 11,400.00 | 2,365.00 | 15,335.00 | | 5.12 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 40.00 | 320.00 | 60.00 | 264.00 | 0.00 |
| Total actual harvest content | 40.00 | 264.48 | 54.87 | 355.77 | 1,187.84 |

3

Field name: 3

04/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 04/09/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 11/29/2023 | 3,200.00 *ton* | Dry-weight | | 71.3 | 11,975.00 | 2,675.00 | 168,000.00 | | 4.67 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 40.00 | 320.00 | 60.00 | 264.00 | 0.00 |
| Total actual harvest content | 40.00 | 274.95 | 61.42 | 3,857.28 | 1,072.23 |

4

Field name: 4

4

04/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 04/10/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 11/29/2023 | 800.00 *ton* | Dry-weight | | 68.0 | 13,850.00 | 1,895.00 | 935.00 | | 5.07 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 40.00 | 320.00 | 60.00 | 264.00 | 0.00 |
| Total actual harvest content | 40.00 | 354.56 | 48.51 | 23.94 | 1,297.92 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

1 - 04/06/2023: Corn, silage

Field name: 1

Crop: Corn, silage

Plant date: 04/06/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/06/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| cover crop plow down | Plowdown credit | 30.00 | 4.80 | 24.90 | 0.00 | |
| Application event totals | | 30.00 | 4.80 | 24.90 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| wastewater application | Process wastewater | 5.35 | 1.62 | 30.04 | 140.99 | 236,842.00 *gal* |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,851,170.00 *gal* |
| Application event totals | | 55.35 | 1.62 | 30.04 | 140.99 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| wastewater application | Process wastewater | 5.35 | 1.62 | 30.04 | 140.99 | 236,842.00 *gal* |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,851,170.00 *gal* |
| Application event totals | | 55.35 | 1.62 | 30.04 | 140.99 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,851,170.00 *gal* |
| Application event totals | | 50.00 | 0.00 | 0.00 | 0.00 | |

## 1 - 04/06/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,932,362.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,688,246.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 3,014,094.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 3,014,094.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,688,246.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 08/04/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| dry manure application | Separator solids | 144.05 | 23.37 | 69.58 | 5,481.30 | 150.00 *ton* |
| Application event totals | | 144.05 | 23.37 | 69.58 | 5,481.30 | |

## 1 - 04/06/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,851,170.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/16/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| UAN 32 | Liquid commercial fertilizer | 10.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 10.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,851,170.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,769,708.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/15/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,199,474.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

## 2 - 04/08/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 04/08/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

## 2 - 04/08/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/06/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| cover crop plowdown credit | Plowdown credit | 30.00 | 4.80 | 24.90 | 0.00 | |
| Application event totals | | 30.00 | 4.80 | 24.90 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| wastewater application | Process wastewater | 5.35 | 1.62 | 30.04 | 140.99 | 631,578.00 *gal* |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,603,120.00 *gal* |
| Application event totals | | 55.35 | 1.62 | 30.04 | 140.99 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| wastewater application | Process wastewater | 5.35 | 1.62 | 30.04 | 140.99 | 631,578.00 *gal* |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,603,120.00 *gal* |
| Application event totals | | 55.35 | 1.62 | 30.04 | 140.99 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,603,120.00 *gal* |
| Application event totals | | 50.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,820,352.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

## 2 - 04/08/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,168,656.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 8,037,584.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 8,037,584.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,168,656.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/04/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| dry manure application | Separator solids | 144.05 | 23.37 | 69.58 | 5,481.30 | 400.00 *ton* |
| Application event totals | | 144.05 | 23.37 | 69.58 | 5,481.30 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,603,120.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

## 2 - 04/08/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/16/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| UAN 32 | Liquid commercial fertilizer | 10.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 10.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,603,120.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,385,888.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/15/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 5,865,264.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

## 3 - 04/09/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 04/09/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/06/2023 | Plow/disc | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| cover crop plowdown credit | Plowdown credit | 30.00 | 4.80 | 24.90 | 0.00 | |
| Application event totals | | 30.00 | 4.80 | 24.90 | 0.00 | |

3 - 04/09/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| wastewater application | Process wastewater | 5.35 | 1.62 | 30.04 | 140.99 | 631,578.00 *gal* |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,603,120.00 *gal* |
| Application event totals | | 55.35 | 1.62 | 30.04 | 140.99 | |

| | | | | |
|---|---|---|---|---|
| 05/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| wastewater application | Process wastewater | 5.35 | 1.62 | 30.04 | 140.99 | 631,578.00 *gal* |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,603,120.00 *gal* |
| Application event totals | | 55.35 | 1.62 | 30.04 | 140.99 | |

| | | | | |
|---|---|---|---|---|
| 06/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,603,120.00 *gal* |
| Application event totals | | 50.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,820,352.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,168,656.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

3 - 04/09/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 8,037,584.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 8,037,584.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,168,656.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 08/05/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| dry manure application | Separator solids | 144.05 | 23.37 | 69.58 | 5,481.30 | 400.00 *ton* |
| Application event totals | | 144.05 | 23.37 | 69.58 | 5,481.30 | |

| | | | | |
|---|---|---|---|---|
| 08/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,603,120.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 08/16/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| UAN 32 | Liquid commercial fertilizer | 10.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 10.00 | 0.00 | 0.00 | 0.00 | |

## 3 - 04/09/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,603,120.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 7,385,888.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 5,865,264.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

## 4 - 04/10/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 04/10/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/06/2023 | Plow/disc | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| cover crop plowdown credit | Plowdown credit | 30.00 | 4.80 | 24.90 | 0.00 | |
| 28-0-0 | Liquid commercial fertilizer | 20.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 50.00 | 4.80 | 24.90 | 0.00 | |

4 - 04/10/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 1,900,780.00 *gal* |
| Application event totals | | 50.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 05/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 1,900,780.00 *gal* |
| Application event totals | | 50.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Liquid commercial fertilizer | 50.00 | 0.00 | 0.00 | 0.00 | |
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 1,900,780.00 *gal* |
| Application event totals | | 50.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 1,955,088.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 1,792,164.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

## 4 - 04/10/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,009,396.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 2,009,396.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 1,792,164.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/04/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| dry manure application | Separator solids | 144.05 | 23.37 | 69.58 | 5,481.30 | 100.00 *ton* |
| Application event totals | | 144.05 | 23.37 | 69.58 | 5,481.30 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 1,900,780.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/16/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| UAN 32 | Liquid commercial fertilizer | 20.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 20.00 | 0.00 | 0.00 | 0.00 | |

4 - 04/10/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 1,900,780.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 09/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 1,846,472.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

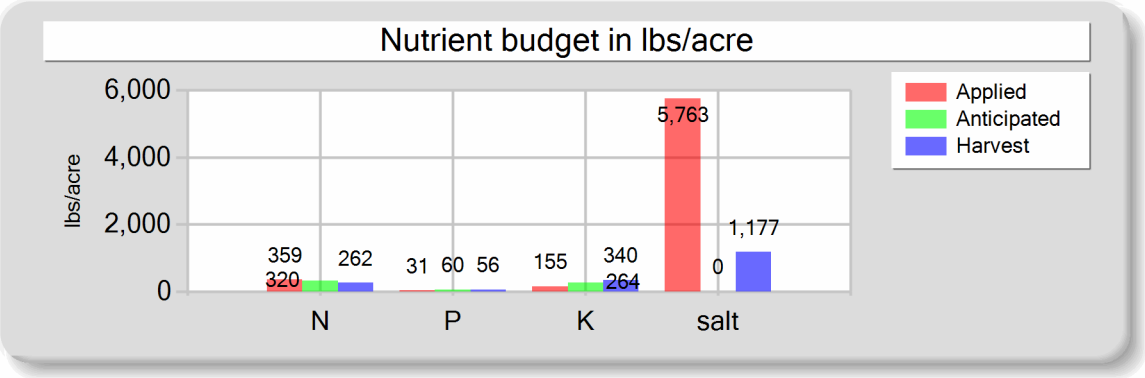
| | | | | |
|---|---|---|---|---|
| 10/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| TID canal | Surface water | 0.00 | 0.00 | 0.00 | 0.00 | 1,466,316.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 0.00 | |

## B. NUTRIENT BUDGET

1 - 04/06/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 04/06/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 30.00 | 4.80 | 24.90 | 0.00 |
| Commercial fertilizer / Other | 160.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 144.05 | 23.37 | 69.58 | 5,481.30 |
| Process wastewater | 10.70 | 3.24 | 60.08 | 281.97 |
| Fresh water | 0.00 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 358.75 | 31.42 | 154.56 | 5,763.27 |
| Anticipated crop nutrient removal | 320.00 | 60.00 | 264.00 | 0.00 |
| Actual crop nutrient removal | 262.33 | 56.48 | 340.38 | 1,177.22 |
| Nutrient balance | 96.43 | -25.06 | -185.81 | 4,586.05 |
| Applied to removed ratio | 1.37 | 0.56 | 0.45 | 4.90 |

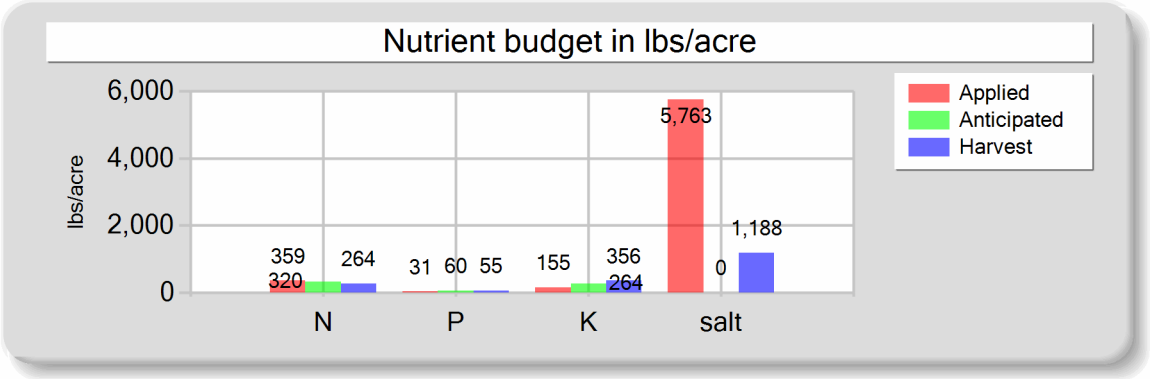
| Fresh water applied | |
|---|---|
| 33,562,074.00 | *gallons* |
| 1,235.98 | *acre-inches* |
| 41.20 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 473,684.00 | *gallons* |
| 17.44 | *acre-inches* |
| 0.58 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## 2 - 04/08/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 04/08/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 30.00 | 4.80 | 24.90 | 0.00 |
| Commercial fertilizer / Other | 160.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 144.05 | 23.37 | 69.58 | 5,481.30 |
| Process wastewater | 10.70 | 3.24 | 60.08 | 281.97 |
| Fresh water | 0.00 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 358.75 | 31.42 | 154.56 | 5,763.27 |
| Anticipated crop nutrient removal | 320.00 | 60.00 | 264.00 | 0.00 |
| Actual crop nutrient removal | 264.48 | 54.87 | 355.77 | 1,187.84 |
| Nutrient balance | 94.27 | -23.45 | -201.21 | 4,575.43 |
| Applied to removed ratio | 1.36 | 0.57 | 0.43 | 4.85 |

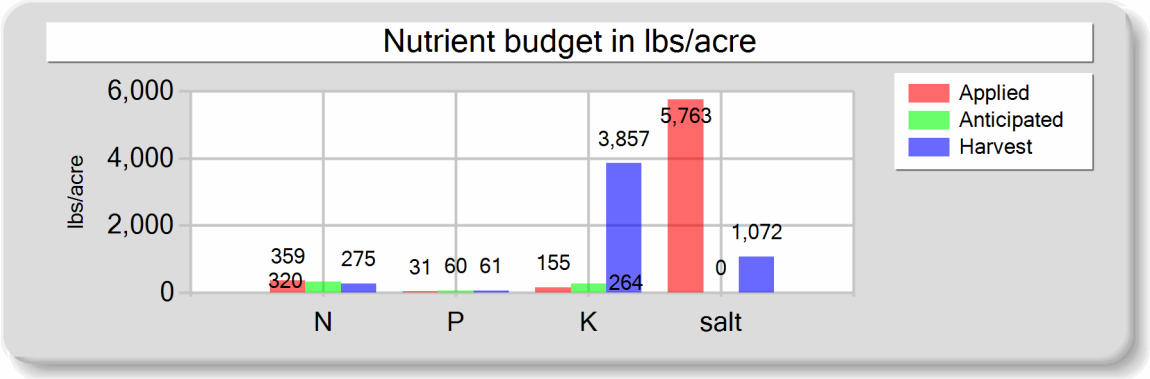
| Fresh water applied | |
|---|---|
| 89,499,584.00 | *gallons* |
| 3,295.97 | *acre-inches* |
| 41.20 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,263,156.00 | *gallons* |
| 46.52 | *acre-inches* |
| 0.58 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## 3 - 04/09/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 04/09/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 30.00 | 4.80 | 24.90 | 0.00 |
| Commercial fertilizer / Other | 160.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 144.05 | 23.37 | 69.58 | 5,481.30 |
| Process wastewater | 10.70 | 3.24 | 60.08 | 281.97 |
| Fresh water | 0.00 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 358.75 | 31.42 | 154.56 | 5,763.27 |
| Anticipated crop nutrient removal | 320.00 | 60.00 | 264.00 | 0.00 |
| Actual crop nutrient removal | 274.95 | 61.42 | 3,857.28 | 1,072.23 |
| Nutrient balance | 83.81 | -30.00 | -3,702.72 | 4,691.04 |
| Applied to removed ratio | 1.30 | 0.51 | 0.04 | 5.38 |

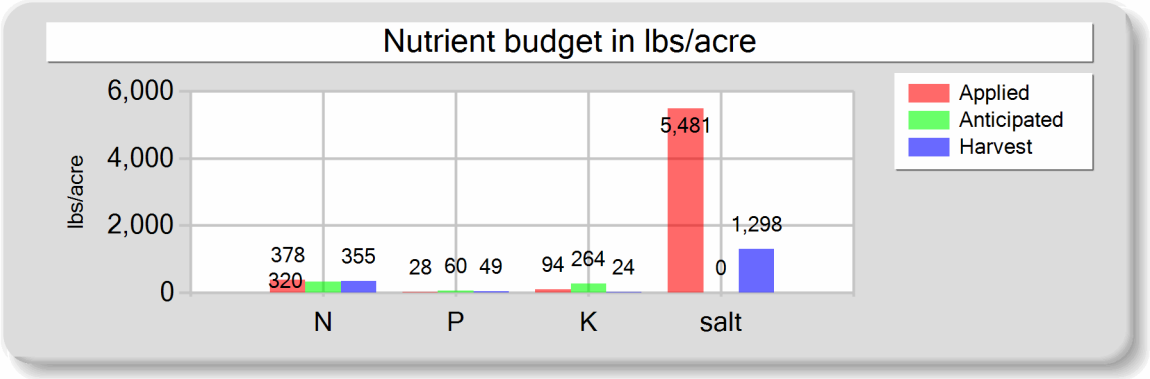
| Fresh water applied | |
|---|---|
| 89,499,584.00 | *gallons* |
| 3,295.97 | *acre-inches* |
| 41.20 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,263,156.00 | *gallons* |
| 46.52 | *acre-inches* |
| 0.58 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## 4 - 04/10/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 04/10/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 30.00 | 4.80 | 24.90 | 0.00 |
| Commercial fertilizer / Other | 190.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 144.05 | 23.37 | 69.58 | 5,481.30 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 0.00 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 378.05 | 28.17 | 94.48 | 5,481.30 |
| Anticipated crop nutrient removal | 320.00 | 60.00 | 264.00 | 0.00 |
| Actual crop nutrient removal | 354.56 | 48.51 | 23.94 | 1,297.92 |
| Nutrient balance | 23.49 | -20.34 | 70.54 | 4,183.38 |
| Applied to removed ratio | 1.07 | 0.58 | 3.95 | 4.22 |

| Fresh water applied | |
|---|---|
| 22,374,896.00 | *gallons* |
| 823.99 | *acre-inches* |
| 41.20 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

t3223005a

Sample and source description: t3223005a

Sample date: 08/11/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 15,900.00 | 2,580.00 | 7,680.00 | 27,100.00 | 5,330.00 | 1,650.00 | 2,130.00 | 1,640.00 | | 60.50 |
| **DL** | 100.00 | 10.00 | 10.00 | 10.00 | 10.00 | 10.00 | 100.00 | 100.00 | | 0.10 |

## B. PROCESS WASTEWATER ANALYSES

t3060006a

Sample and source description: t3060006a

Sample date: 03/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 269.00 | 171.00 | 0.00 | 0.00 | 48.10 | 474.00 | | | | | | | | 3,880.00 | 3,580 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

t3188002a late 2nd quarter

Sample and source description: t3188002a late 2nd quarter

Sample date: 07/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 81.20 | 8.40 | 0.00 | 0.00 | 24.60 | 456.00 | | | | | | | | 2,590.00 | 2,140 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

## C. FRESH WATER ANALYSES

TID canal

TID canal

t3214016a

Sample description: t3214016a

Sample date: 08/02/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 0.00 | 0.00 | 0.00 | | | | | | | | 0.00 | 0 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

## D. SOIL ANALYSES

*No soil analyses entered.*

## E. PLANT TISSUE ANALYSES

1 - 04/06/2023: Corn, silage

t3216005a, t3334033c

Sample and source description: t3216005a, t3334033c

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,100.00 | 2,605.00 | 15,700.00 | | 5.43 |
| **DL** | 100.00 | 10.00 | 10.00 | | 0.10 |

2 - 04/08/2023: Corn, silage

## 2 - 04/08/2023: Corn, silage

### t3214027a, t3216004a, t3334033a, t3333018b

Sample and source description: t3214027a, t3216004a, t3334033a, t3333018b

Sample date: 11/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,400.00 | 2,365.00 | 15,335.00 | | 5.12 |
| **DL** | 100.00 | 10.00 | 10.00 | | 0.10 |

## 3 - 04/09/2023: Corn, silage

### t3214027b, t3216004b,t3333018c,t3334033b

Sample and source description: t3214027b, t3216004b,t3333018c,t3334033b

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,975.00 | 2,675.00 | 168,000.00 | | 4.67 |
| **DL** | 100.00 | 10.00 | 10.00 | | 0.10 |

## 4 - 04/10/2023: Corn, silage

### t3214026a, t3333018a

Sample and source description: t3214026a, t3333018a

Sample date: 11/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 13,850.00 | 1,895.00 | 935.00 | | 5.07 |
| **DL** | 100.00 | 10.00 | 10.00 | | 0.10 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

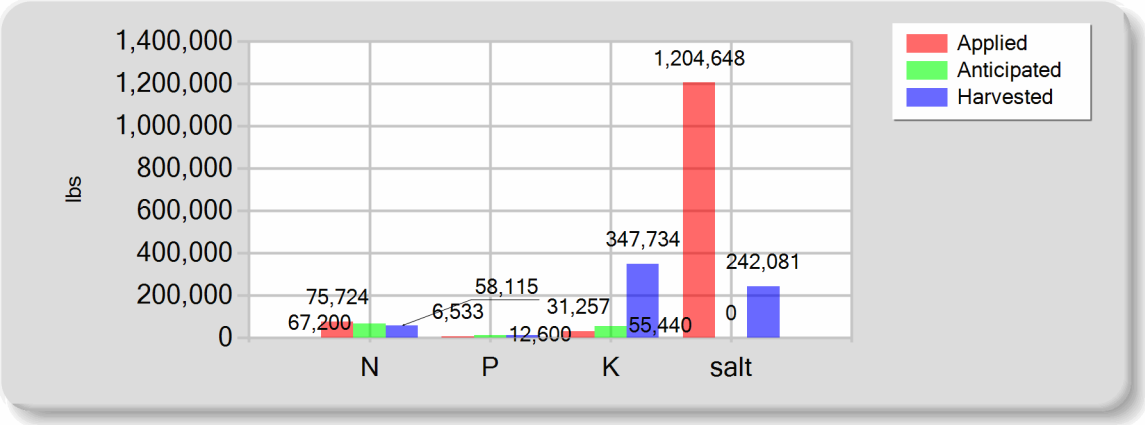
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

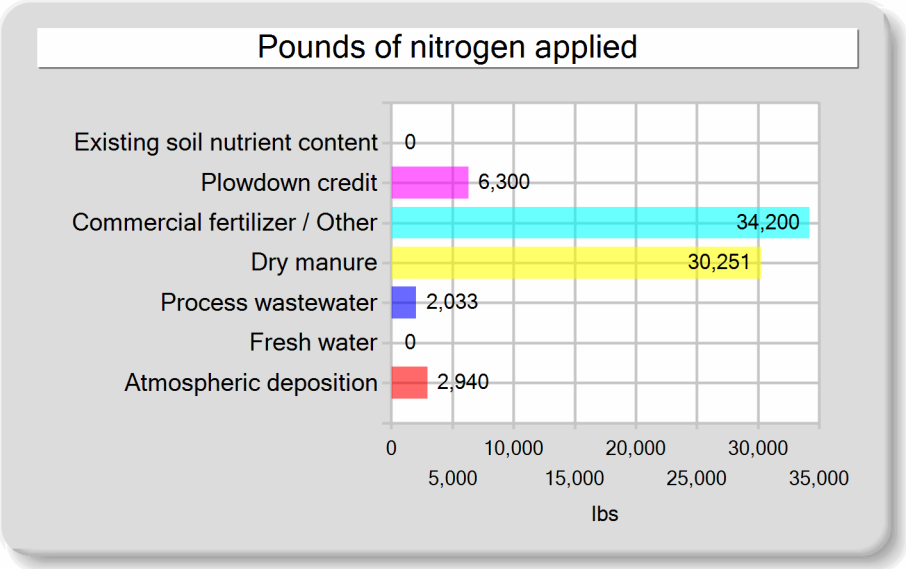
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

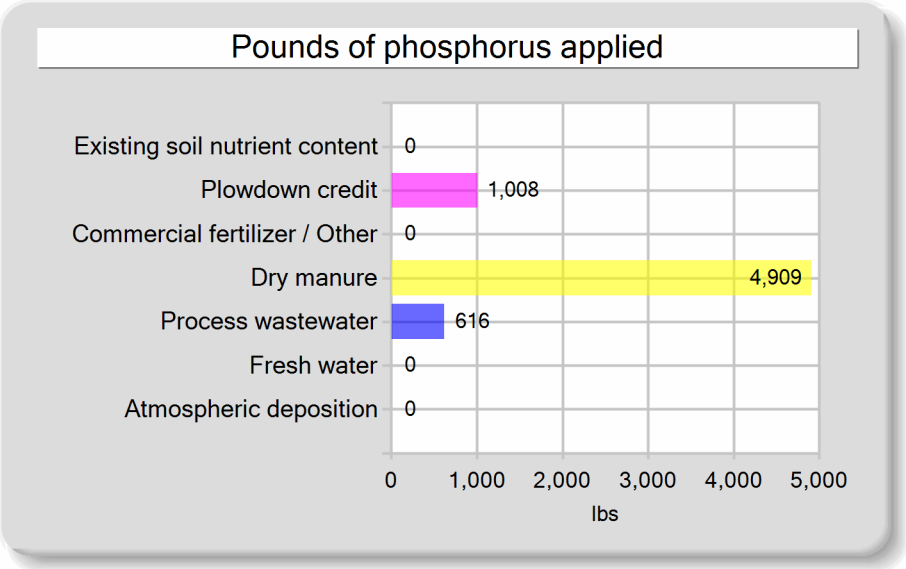
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 6,300.00 | 1,008.00 | 5,229.00 | 0.00 |
| Commercial fertilizer / Other | 34,200.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 30,251.34 | 4,908.71 | 14,611.97 | 1,151,073.00 |
| Process wastewater | 2,032.84 | 615.86 | 11,415.94 | 53,574.83 |
| Fresh water | 0.00 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 2,940.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 75,724.18 | 6,532.57 | 31,256.91 | 1,204,647.83 |
| Anticipated crop nutrient removal | 67,200.00 | 12,600.00 | 55,440.00 | 0.00 |
| Actual crop nutrient removal | 58,115.12 | 11,967.41 | 347,734.16 | 242,080.88 |
| Nutrient balance | 17,609.06 | -5,434.84 | -316,477.25 | 962,566.95 |
| Applied to removed ratio | 1.30 | 0.55 | 0.09 | 4.98 |

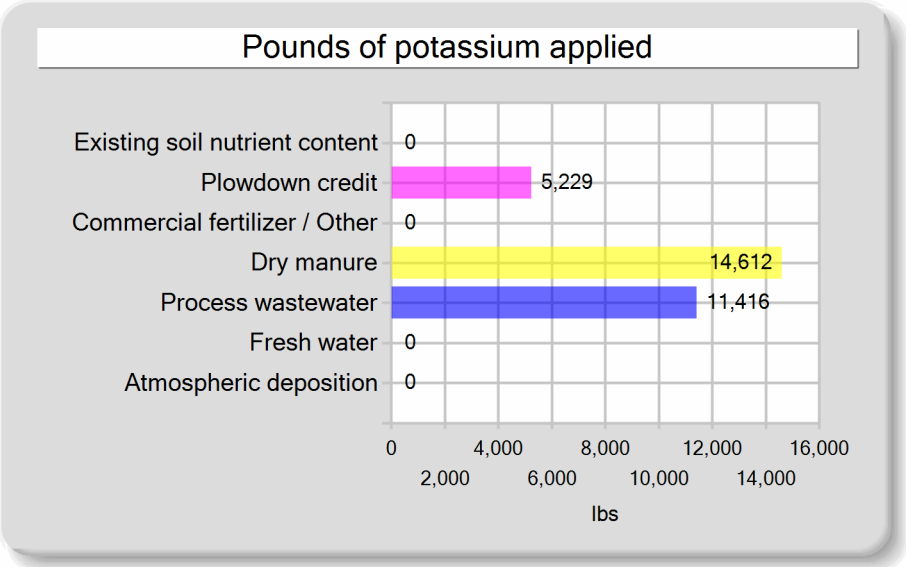
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

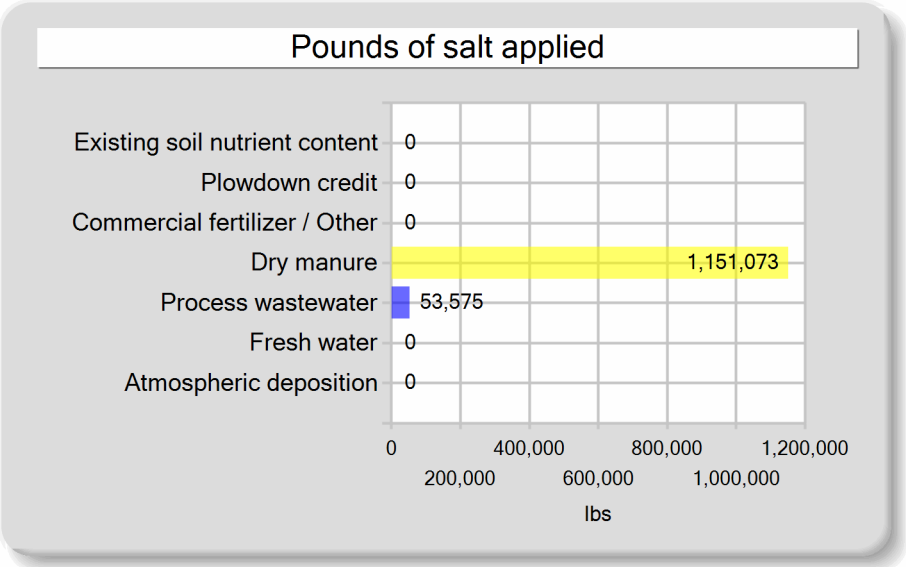


## C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | Yes |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

ADDITIONAL NOTES

**A. NOTES**

Winter forage crop was terminated in March for early planting of double crop corn. Termination was at approximately 10". Plow down credit applied to corn for this crop. Winter forage was planted 12/1/2023.
Cows were sold May 2023. Facility has been empty since this time. Since the facility was empty over half of the year, the animal numbers were reduced to half of the beginning population.
Last import date of commercial fertilizer for NH3 was entered for import date for all applications.
Harvest date of 2nd crop of corn was used for harvest date.
Plant tissue samples were taken on a 40 acres basis.
Corn samples were averaged for 1st and 2nd crop of submittal purposes.
Deep Wells were not sampled as they were never used during 2023 and it is to costly to run them for purge purposes only.

CERTIFICATION

**A. OWNER AND/OR OPERATOR CERTIFICATION**

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
|---|---|
| Adam Crowell | Dairy Bar Vee |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| | |
| DATE | DATE |

ATTACHMENTS

## A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

| Annual Dairy Facility Assessment |
| --- |
| Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A ) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed . |

| Manure/Process Wastewater Tracking Manifests |
| --- |
| Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler. |

| Corrective Actions Documents |
| --- |
| Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction. |

| Groundwater Monitoring |
| --- |
| Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13. |

| Storm Water Monitoring |
| --- |
| Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14. |

CERTIFICATION

## A. OWNER AND/OR OPERATOR CERTIFICATION

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| Adam Crowell | Dairy Bar Vee |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| 6/13/24 | 6/13/24 |
| DATE | DATE |

**Denele Analytical, Inc.**

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

# Compliance Analysis Report

**Customer**
BAR VEE DAIRY
P.O. Box 3088
Turlock, CA 95381

**Grower**

**Consultant:**
CDC

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:50 AM
SUBMITTED BY: Adam
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.2c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| Domestic Well A | T3362022A | Electrical Conductivity (EC) | EPA 120.1 and | 1.14 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 36.5 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 730 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 91.0 | | | |
| Electrical Conductivity (EC) | 95.4 | | | 0.677 |
| Nitrate Nitrogen (NO3-N) | 105 | 99.5 (T3362016J) | 98.2 (T3362016J) | 1.25 |

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.



ELAP Certificate No.: 2714

# DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

## DAIRY COMPLIANCE

Client: Bar Vee Dairy

Mailing address:
(not facility address)

Phone#:

Authorized Copy Released To: CDC

Sampled By:

Project ID:

## ANALYSIS TO BE COMPLETED:

**IRRIGATION/GROUND WATER**
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

**PROCESS WASTE WATER**
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

**TILE DRAIN**
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

**DISCHARGE WATER**
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

**MANURE**
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

**PLANT TISSUE**
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

**SOIL**
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

| | Date Collected | Time Collected | Sample ID | Description | Temp | Well Purge | Analysis: H1 | H3 | M3 | F18 | OTHER |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 12/27/23 | 3pm | Domestic Wells | A | 20.2°C | | ✓ | | | | |
| 2 | | | | | | | | | | | |
| 3 | | | | | | | | | | | |
| 4 | | | | | | | | | | | |
| 5 | | | | | | | | | | | |
| 6 | | | | | | | | | | | |
| 7 | | | | | | | | | | | |
| 8 | | | | | | | | | | | |
| 9 | | | | | | | | | | | |
| 10 | | | | | | | | | | | |

**CHAIN OF CUSTODY RECORDING**

| | Signature | Company | Received Date & Time | Relinquished Date & Time | *(LABORATORY USE ONLY)* Logged in By: | LABORATORY NUMBER |
|---|---|---|---|---|---|---|
| 1st | | | | | | |
| 2nd | | DENELE ANALYTICAL, INC. | 12-28-23 @ 11:50 | | | T3362022 |

Required to be submitted to any agency? (YES / NO)

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# Compliance Analysis Report

**Customer**
BAR VEE DAIRY
P.O. Box 3088
Turlock, CA 95381

**Grower**

**Consultant:**
CDC

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:50 AM
SUBMITTED BY: Adam
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.0c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| Domestic Well B | T3362023A | Electrical Conductivity (EC) | EPA 120.1 and | 0.634 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 11.2 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 405 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 91.0 | | | |
| Electrical Conductivity (EC) | 95.4 | | | 0.677 |
| Nitrate Nitrogen (NO3-N) | 105 | 99.5 (T3362016J) | 98.2 (T3362016J) | 1.25 |

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PH# 209-634-9055 FAX# 209-634-9057

## DAIRY COMPLIANCE

Client: Bar Vee Dairy

Mailing address:
(not facility address)

Phone#:

Authorized Copy Released To: CDC

Sampled By:

Project ID:

## ANALYSIS TO BE COMPLETED:

**IRRIGATION/GROUND WATER**
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

**PROCESS WASTE WATER**
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

**TILE DRAIN**
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

**DISCHARGE WATER**
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

**MANURE**
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

**PLANT TISSUE**
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

**SOIL**
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

| | Date Collected | Time Collected | Sample ID | Description | Temp | Well Purge | H1 | H3 | M3 | F18 | OTHER |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 12/27/23 | 3pm | Domestic Well B | | 20.0°C | | ✓ | | | | |
| 2 | | | | | | | | | | | |
| 3 | | | | | | | | | | | |
| 4 | | | | | | | | | | | |
| 5 | | | | | | | | | | | |
| 6 | | | | | | | | | | | |
| 7 | | | | | | | | | | | |
| 8 | | | | | | | | | | | |
| 9 | | | | | | | | | | | |
| 10 | | | | | | | | | | | |

Analysis

## CHAIN OF CUSTODY RECORDING

| | Signature | Company | Received Date & Time | Relinquished Date & Time | (LABORATORY USE ONLY) Logged in By: | LABORATORY NUMBER |
|---|---|---|---|---|---|---|
| 1st | | | | | | |
| 2nd | | DENELE ANALYTICAL, INC. | 12-28-23 @ 11:50 | | | T3362023 |

Required to be submitted to any agency? (YES / NO)